

Multiple Choice (10 marks)

1. The eigenvalues of a Hermitian operator are always
 - (a) real
 - (b) imaginary
 - (c) degenerate
 - (d) linear
2. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
3. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
4. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic
5. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
 - (a) 44 Hz
 - (b) 196 Hz
 - (c) 262 Hz
 - (d) 393 Hz

True / False (10 marks)

Answer True or False

6. True or False: For a single-electron atom in the $l = 2$ state, there are only 3 allowed values of the magnetic quantum number m_l .
7. True or False: Electromagnetic radiation emitted from a nucleus is most commonly in the form of gamma rays.
8. True or False: A dye laser is the best type of laser for performing spectroscopy across a broad range of visible wavelengths.
9. True or False: A particle moving at $0.60c$ will travel only 288 m before decaying, based on its rest frame decay time of 2.0 ms.
10. True or False: The temperature of a system undergoing a reversible process must change throughout the process.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between the kinetic energy of electrons and their ability to produce a diffraction pattern when interacting with a crystal lattice, specifically considering a lattice spacing of 0.4 nm. What is the approximate energy required, and why is this significant in the context of electron diffraction?
12. Discuss the relationship between linear momentum and wavelength of a free massive particle as proposed by De Broglie, highlighting the role of Planck's constant in this context.

End of Paper